

Chapter 4

Continuous-Time Markov Chain

4.1 Continuous-Time Markov Chains

DEF 4.1.1 (CONTINUOUS-TIME MARKOV CHAIN)

The process $\{X(t), t \geq 0\}$ is a continuous-time Markov chain (CTMC) if for all $s, t \geq 0$ and nonnegative integers $i, j, x(u), 0 \leq u < s$

$$P[X(s+t) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s] = P[X(s+t) = j \mid X(s) = i]$$

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In other words, a CTMC is a sp having the Markovian property that the conditional dist. of the future $X(t+s)$ given the present $X(s)$ and the past $X(u), 0 \leq u < s$, depends only on the present and is independent of the past. If, in addition, $P[X(s+t) = j \mid X(s) = i]$ is independent of s , then the CTMC is said to have stationary or homogeneous transition probabilities.

Ex 4.1.1 In CTMC, let's suppose that the process enters state i at 0 and suppose that the process does not leave state i during the next 10 minutes. What is the prob. that process will not leave state i during the following 5 min?

Sol Let T_i denote the amount of time that the process stays in state i before making a transition.

$$P[T_i > 15 \mid T_i > 10] = P[T_i > 5]$$

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In general,

$$P[T_i > s+t \mid T_i > s] = P[T_i > t]$$

for all $s, t \geq 0$. Hence, the rv T_i is memoryless and must thus be exponentially distributed.

From this, we can give another definition of CTMC.

DEF 4.1.2 (CTMC)

The CTMC is a sp having the property that each time it enters state i

- i. the amount of time it spends in that state before making a transition into a different state is exponentially distributed with mean, say, $1/v_i$, and
- ii. when the process leave state i , it next enters state j with some probability, say, P_{ij} . Of course, the P_{ij} must satisfy

$$\begin{aligned} P_{ii} &= 0, & \text{all } i \\ \sum_j P_{ij} &= 1, & \text{all } i \end{aligned}$$

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In other words, a CTMC is a sp that moves from state to state in accordance with a Markov chain, but is such that the amount of time it spends in each state, before proceeding to the next state, is exponentially distributed.

EX 4.1.2 Consider a shoeshine shop consisting of 2 chairs—chair 1 and chair 2. A customer upon arrival goes initially to chair 1 and then to chair 2. The service times at two chairs are assumed to be independent rvs which are exponentially distributed with service rates μ_1 and μ_2 . Suppose that potential customers arrive in accordance with a Poisson process having rate λ , and that a potential customers will only enter the system if both chairs are empty. Model the system state by CTMC.

Sol: $v_0 = \lambda, v_1 = \mu_1, v_2 = \mu_2. P_{01} = P_{12} = P_{20} = 1$

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4.2 Birth and Death Processes

DEF 4.2.1

A birth and death process is a CTMC with states $\{0, 1, \dots\}$ for which transitions from state n may go only to either state $n - 1$ or state $n + 1$.

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EX 4.2.1 Suppose there are n people in the system, then

- i. new arrivals enter the system at an exponential rate λ_n , and
- ii. people leave the system at an exponential rate μ_n .

This process is a birth and death process and the state transition rates and probabilities are

$$\begin{aligned} v_0 &= \lambda_0, \\ v_i &= \lambda_i + \mu_i, \quad i > 0 \\ P_{01} &= 1, \\ P_{i,i+1} &= \frac{\lambda_i}{\lambda_i + \mu_i}, \quad i > 0 \\ P_{i,i-1} &= \frac{\mu_i}{\lambda_i + \mu_i}, \quad i > 0 \end{aligned}$$

The parameters $\{\lambda_0, \lambda_1, \lambda_2, \dots\}$ and $\{\mu_0, \mu_1, \mu_2, \dots\}$ are called the arrival (or birth) and departure (or death) rates, respectively.

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Ex 4.2.2 Consider a birth and death process for which

$$\begin{aligned} \mu_n &= 0, & \text{for all } n \geq 0 \\ \lambda_n &= \lambda, & \text{for all } n \geq 0 \end{aligned}$$

This is a process in which departures never occur, and the time between successive arrivals is exponential with mean $1/\lambda$. Hence, this is just the Poisson process.

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Ex 4.2.3 (M/M/1 QUEUEING SYSTEM) Suppose that customers arrive at a single-server service station with a Poisson process having rate λ . Upon arrival, each customer goes directly into service if the server is free; if not, then the customer joins the queue. When the server finishes serving a customer, the customer leaves the system and the next customer in line, if there are any waiting, enters the service. This is known as M/M/1 queueing system.

If we let $X(t)$ denote the number of customers in the system at time t then $\{X(t), t \geq 0\}$ is a birth and death process with

$$\begin{aligned} \mu_n &= \mu, & n \geq 1 \\ \lambda_n &= \lambda, & n \geq 0 \end{aligned}$$

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Ex 4.2.4 (A MULTISERVER EXPONENTIAL QUEUEING SYSTEM) Consider an exponential queueing system in which there are s servers available, each server at rate μ . An entering customer first waits in line and then goes to the first free server. This is a birth and death process with parameters

$$\mu_n = \begin{cases} n\mu, & 1 \leq n \leq s \\ s\mu, & n > s \end{cases}$$

$$\lambda_n = \lambda, \quad n \geq 0$$

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4.3 The Transition Probability Function

DEF 4.3.1

Transition probability is defined as the probability that a process presently in state i will be in state j a time t later:

$$P_{ij}(t) = P[X(t+s) = j \mid X(s) = i]$$

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To determine $P_{ij}(t)$ in the case of a pure birth process, let X_k denote the time the process spends in state k before making a transition into state $k+1$, $k \geq 1$. Since

$$X(t) < j \Leftrightarrow X_i + \cdots + X_{j-1} > t,$$

$$P[X(t) < j \mid X(0) = i] = P\left[\sum_{k=i}^{j-1} X_k > t\right]$$

Since $X_i, \dots, X_{j-1}$ are independent exponential rvs with respective rates $\lambda_i, \dots, \lambda_{j-1}$, we obtain

$$P[X(t) < j \mid X(0) = i] = \sum_{k=i}^{j-1} e^{-\lambda_k t} \prod_{r \neq k, r=i}^{j-1} \frac{\lambda_r}{\lambda_r - \lambda_k}$$

(Hypoexponential, Refer to 5.9 in pp. 356)

Since

$$P[X(t) = j \mid X(0) = i] = P[X(t) < j+1 \mid X(0) = i] - P[X(t) < j \mid X(0) = i]$$

we have shown the following.

PROPOSITION 4.3.2

For a pure birth process having $\lambda_i \neq \lambda_j$ when $i \neq j$

$$P_{ij}(t) = \sum_{k=i}^j e^{-\lambda_k t} \prod_{r \neq k, r=i}^j \frac{\lambda_r}{\lambda_r - \lambda_k} - \sum_{k=i}^{j-1} e^{-\lambda_k t} \prod_{r \neq k, r=i}^{j-1} \frac{\lambda_r}{\lambda_r - \lambda_k}, \quad i < j \quad (4.1)$$

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For any pair of states i and j , let

$$q_{ij} = v_i P_{ij} \quad (4.2)$$

Since v_i is the rate at which the process make a transition when in state i and P_{ij} is the probability that this transition is into state j , it follows that q_{ij} is the rate, when in state i , at which the process makes a transition into state j . The quantities q_{ij} are called the instantaneous transition rates. Since

$$v_i = \sum_j v_i P_{ij} = \sum_j q_{ij} \quad (4.3)$$

and

$$P_{ij} = \frac{q_{ij}}{v_i} = \frac{q_{ij}}{\sum_j q_{ij}} \quad (4.4)$$

it follows that specifying the instantaneous transition rates determines the parameters of the CTMC. Note that CTMC was specified by v_i and P_{ij} .

LEMMA 4.3.3

i.

$$\lim_{h \rightarrow 0} \frac{1 - P_{ii}(h)}{h} = v_i \quad (4.5)$$

ii.

$$\lim_{h \rightarrow 0} \frac{P_{ij}(h)}{h} = q_{ij}, \quad \text{when } i \neq j \quad (4.6)$$

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Using the following facts we can prove the lemmas.

$$\begin{aligned} P[t \leq t_0 + h \mid t > t_0] &= 1 - e^{\lambda h} = 1 - (1 - \lambda h + \frac{(\lambda h)^2}{2!} - \dots) \\ &= \lambda h + o(h) \end{aligned}$$

$$\begin{aligned} P[t > t_0 + h \mid t > t_0] &= 1 - (1 - e^{\lambda h}) = e^{-\lambda h} = 1 - \lambda h + \frac{(\lambda h)^2}{2!} - \dots \\ &= 1 - \lambda h + o(h) \end{aligned}$$

$$\begin{aligned} P[2 \text{ or more events in } (t_0, t_0 + h)] &= 1 - \Pr[\text{no events}] - \Pr[1 \text{ event in } (t_0, t_0 + h)] \\ &= 1 - (1 - \lambda h + o(h)) - (\lambda h + o(h)) \\ &= o(h) \end{aligned}$$

From these,

$$\begin{aligned} 1 - P_{ii}(h) &= 1 - \Pr[\text{no transition during time } h] \\ &= \lambda h + o(h) \end{aligned}$$

Dividing by h and taking the limit over h , we obtain the 1st lemma. Similarly, we have

$$\begin{aligned} P_{ij}(h) &= \Pr[\text{transition from } i \text{ to } j \text{ during time } h] \\ &= \Pr[1 \text{ transition during time } h]P_{ij} + \Pr[2 \text{ transitions or more during time } h]P_{ij} \\ &= hv_i P_{ij} + o(h) \end{aligned}$$

Dividing by h and taking the limit over h , we obtain the 2nd lemma.

LEMMA 4.3.4 (CHAPMAN-KOLMOGOROV EQUATIONS)

For all $s \geq 0, t \geq 0$,

$$P_{ij}(t+s) = \sum_{k=0}^{\infty} P_{ik}(t)P_{kj}(s) \quad (4.7)$$

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Proof:

$$\begin{aligned} P_{ij}(t+s) &= P[X(t+s) = j \mid X(0) = i] \\ &= \sum_{k=0}^{\infty} P[X(t+s) = j, X(t) = k \mid X(0) = i] \\ &= \sum_{k=0}^{\infty} P[X(t+s) = j \mid X(t) = k, X(0) = i]P[X(t) = k \mid X(0) = i] \\ &= \sum_{k=0}^{\infty} P[X(t+s) = j \mid X(t) = k]P[X(t) = k \mid X(0) = i] \\ &= \sum_{k=0}^{\infty} P_{kj}(s)P_{ik}(t) \end{aligned}$$

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From the Lemma,

$$\begin{aligned} P_{ij}(t+h) - P_{ij}(t) &= \sum_{k=0}^{\infty} P_{ik}(h)P_{kj}(t) - P_{ij}(t) \\ &= \sum_{k \neq i} P_{ik}(h)P_{kj}(t) - [1 - P_{ii}(h)]P_{ij}(t) \end{aligned}$$

and thus

$$\lim_{h \rightarrow 0} \frac{P_{ij}(t+h) - P_{ij}(t)}{h} = \lim_{h \rightarrow 0} \left\{ \sum_{k \neq i} \frac{P_{ik}(h)}{h} P_{kj}(t) - \left[\frac{1 - P_{ii}(h)}{h} \right] P_{ij}(t) \right\}$$

We obtain

$$P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i P_{ij}(t)$$

It gives the following theorem.

THEOREM 4.3.5 (KOLMOGOROV'S BACKWARD EQUATION)

For all states i, j and time $t \geq 0$,

$$P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - v_i P_{ij}(t) \quad (4.8)$$

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Ex 4.3.1 The backward equation for the pure birth process become

$$P'_{ij}(t) = \lambda_i P_{i+1,j}(t) - \lambda_i P_{ij}(t) \quad (4.9)$$

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Ex 4.3.2 The backward equation for the pure birth and death process become

$$\begin{aligned} P'_{0j}(t) &= \lambda_0 P_{1,j}(t) - \lambda_0 P_{0j}(t) \\ P'_{ij}(t) &= (\lambda_i + \mu_i) \left[\frac{\lambda_i}{\lambda_i + \mu_i} P_{i+1,j}(t) + \frac{\mu_i}{\lambda_i + \mu_i} P_{i-1,j}(t) \right] - (\lambda_i + \mu_i) P_{ij}(t) \end{aligned}$$

or equivalently,

$$P'_{0j}(t) = \lambda_0 P_{1,j}(t) - \lambda_0 P_{0j}(t) \quad (4.10)$$

$$P'_{ij}(t) = \lambda_i P_{i+1,j}(t) + \mu_i P_{i-1,j}(t) - (\lambda_i + \mu_i) P_{ij}(t), \quad i > 0 \quad (4.11)$$

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Ex 4.3.3 Consider a machine that works for an exponential amount of time having mean $1/\lambda$ before breaking down; and suppose that it takes an exponential amount of time having mean $1/\mu$ to repair the machine. If the machine is in working condition at time 0, then what is the probability that it will be working at time $t = 10$?

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From Lemma 4.3.4, we have

$$\begin{aligned} P_{ij}(t+h) - P_{ij}(t) &= \sum_{k=0}^{\infty} P_{ik}(t) P_{kj}(h) - P_{ij}(t) \\ &= \sum_{k \neq j} P_{ik}(t) P_{kj}(h) - [1 - P_{jj}(h)] P_{ij}(t) \end{aligned}$$

and thus

$$\lim_{h \rightarrow 0} \frac{P_{ij}(t+h) - P_{ij}(t)}{h} = \lim_{h \rightarrow 0} \left\{ \sum_{k \neq j} P_{ik}(t) \frac{P_{kj}(h)}{h} - \left[\frac{1 - P_{jj}(h)}{h} \right] P_{ij}(t) \right\}$$

We obtain

$$P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j P_{ij}(t)$$

THEOREM 4.3.6 (KOLMOGOROV'S FORWARD EQUATION)

Under suitable regularity conditions,

$$P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j P_{ij}(t)$$

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PROPOSITION 4.3.7

For a pure birth process,

$$\begin{aligned} P_{ii}(t) &= e^{-\lambda_i t}, \quad i \geq 0 \\ P_{ij}(t) &= \lambda_{j-1} e^{-\lambda_j t} \int_0^t e^{\lambda_j s} P_{i,j-1}(s) ds, \quad j \geq i+1 \end{aligned}$$

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4.4 Limiting Probabilities

In analogy with DTMC, the probability that a CTMC will be in state j at time t often converges to a limiting value which is independent of the initial state:

$$P_j = \lim_{t \rightarrow \infty} P_{ij}(t)$$

To derive a set of equations for the P_j , consider the set of forward equations

$$P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - v_j P_{ij}(t)$$

Letting t approach ∞ , we have

$$\begin{aligned} \lim_{t \rightarrow \infty} P'_{ij}(t) &= \lim_{t \rightarrow \infty} \left[\sum_{k \neq j} q_{kj} P_{ik}(t) - v_j P_{ij}(t) \right] \\ &= \sum_{k \neq j} q_{kj} P_k - v_j P_j \end{aligned}$$

Since $P'_{ij}(t)$ must converge to 0, we have

$$0 = \sum_{k \neq j} q_{kj} P_k - v_j P_j$$

or

$$v_j P_j = \sum_{k \neq j} q_{kj} P_k, \quad \text{all states } j \quad (4.12)$$

The preceding set of equations, along with this equation,

$$\sum_j P_j = 1 \quad (4.13)$$

can be used to solve the limiting probabilities.

A sufficient condition for existence of limiting probabilities P_j is that

- i. All states of the MC communicate.
- ii. The MC is positive recurrent.

Eq. (4.12) and (4.13) has a nice interpretation:

$$v_j P_j = \text{rate at which the process leaves state } j$$

$$\sum_{k \neq j} q_{kj} P_k = \text{rate at which the process enters state } j$$

Eq. (4.12) is just a statement of the equality of the rates at which the process enters and leaves the state j . Because it balances these rates, the Eq. (4.12) are sometimes referred to as “balance equations.”

Let us now determine the limiting probabilities for a birth and death process. From Eq. (4.12), we obtain

State	Rate at which leave = rate at which enter
0	$\lambda_0 P_0 = \mu_1 P_1$
1	$(\lambda_1 + \mu_1) P_1 = \mu_2 P_2 + \lambda_0 P_0$
2	$(\lambda_2 + \mu_2) P_2 = \mu_3 P_3 + \lambda_1 P_1$
...	...
$n, n \geq 1$	$(\lambda_n + \mu_n) P_n = \mu_{n+1} P_{n+1} + \lambda_{n-1} P_{n-1}$

By adding to each equation the equation preceding it, we obtain

$$\begin{aligned}
 \lambda_0 P_0 &= \mu_1 P_1, \\
 \lambda_1 P_1 &= \mu_2 P_2, \\
 \lambda_2 P_2 &= \mu_3 P_3, \\
 &\dots \\
 \lambda_n P_n &= \mu_{n+1} P_{n+1}, \quad n \geq 0
 \end{aligned}$$

Solving in terms of P_0 yields

$$\begin{aligned}
 P_1 &= \frac{\lambda_0}{\mu_1} P_0, \\
 P_2 &= \frac{\lambda_1}{\mu_2} P_1 = \frac{\lambda_1 \lambda_0}{\mu_2 \mu_1} P_0, \\
 P_3 &= \frac{\lambda_2}{\mu_3} P_2 = \frac{\lambda_2 \lambda_1 \lambda_0}{\mu_3 \mu_2 \mu_1} P_0, \\
 &\dots \\
 P_n &= \frac{\lambda_{n-1}}{\mu_n} P_{n-1} = \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_2 \mu_1} P_0
 \end{aligned}$$

And by using the fact that $\sum_{n=0}^{\infty} P_n = 1$, we obtain

$$1 = P_0 + P_0 \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_2 \mu_1}$$

or

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_2 \mu_1}}$$

And so

$$P_n = \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_2 \mu_1 \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_2 \mu_1} \right)}, \quad n \geq 1 \quad (4.14)$$

From the above, the necessary condition for the limiting probabilities to exist is

$$\sum_{n=1}^{\infty} \frac{\lambda_{n-1} \lambda_{n-2} \cdots \lambda_1 \lambda_0}{\mu_n \mu_{n-1} \cdots \mu_2 \mu_1} < \infty$$

This condition also may be shown to be sufficient.

EX 4.4.1 (EX 6.14, PP. 373) M/M/1

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EX 4.4.2 (EX 6.15, PP. 374) Consider the showshine shop and determine the proportion of the time the process is in each of the states 0, 1, 2. Note that it is not a birth and death process.

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Chapter 5

Queueing Theory

We study a class of models in which customers arrive in some random manner to a service facility. Upon arrival they are made to wait in queue until it is their turn to be served. Once served they are generally assumed to leave the system.

For such models we will be interested in determining such quantities as

- i. the average number of customers in the system (or in the queue) and
- ii. the average amount of time a customer spends in the system (or spends waiting in the queue).

5.1 Little's Theorem

In a general queueing system when the number of customers is large then so is the waiting time. Here we like to establish a very simple relationship between the mean number in the system, the mean arrival rate, and the mean system time.

Let us look at the input of the queueing system and count how many customers enter as a function of time. We denote this by $\alpha(t)$ where

$$\alpha(t) = \text{number of arrivals in } (0, t)$$

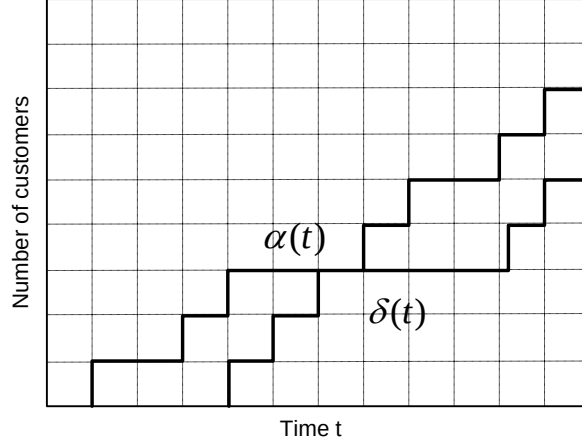
Alternatively, we count the number of departures at the output of the queueing system; we denote this by

$$\delta(t) = \text{number of departures in } (0, t)$$

Clearly $N(t)$, the number in the system at time t , must be given by

$$N(t) = \alpha(t) - \delta(t)$$

The total area between two curves up to some point t represents the total time all customers have spent in the system during the interval $(0, t)$; let us denote this cumulative area by $\gamma(t)$. Moreover let λ_t be



defined as the average arrival rate during the interval $(0, t)$; that is,

$$\lambda_t = \frac{\alpha(t)}{t}$$

We define T_t as the system time per customer averaged over all customers in the interval $(0, t)$; since $\gamma(t)$ represents the accumulated customer-second up to time t , we may divide by the number of arrivals up to that point to obtain

$$T_t = \frac{\gamma(t)}{\alpha(t)}$$

Lastly we define $\bar{N}_t$ as the average number of customers in the queueing system during the interval $(0, t)$; this may be obtained by dividing the accumulated number of customer-seconds by the total interval length t

$$\bar{N}_t = \frac{\gamma(t)}{t}$$

From these last three equations we see

$$\bar{N}_t = \lambda_t T_t$$

Let us now assume that our queueing system is such that the following limits exist as $t \rightarrow \infty$.

$$\begin{aligned} \lambda &= \lim_{t \rightarrow \infty} \lambda_t \\ T &= \lim_{t \rightarrow \infty} T_t \end{aligned}$$

Then,

$$\bar{N} = \lambda T \tag{5.1}$$

This results is known as Little's result. It states that the average number of customers in a queueing system is equal to the average arrival rate of customers to that system, times the average time spent in that system.

Ex 5.1.1 Customers arrive at a fast-food restaurant at a rate of five per minute and wait to receive their order for an average of 5 minutes. Customers eat in the restaurant with prob. 0.5 and carry out their order with prob. 0.5. A meal requires an average of 20 minutes. What is the average number of customers in the restaurant? Ans: 75

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Ex 5.1.2 Consider a network of transmission lines where packets arrive at n different nodes with corresponding rates $\lambda_1, \dots, \lambda_n$. If N is the average total number of packets inside the network find the average delay per packet.

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5.2 The M/M/1 Systems

The M/M/1 queueing system consists of a single queueing system with a single server. Customer arrive according to a Poisson process with rate λ , and the pdf of the service time is exponential with mean $1/\mu$ sec. The name M/M/1 reflects standard queueing theory nomenclature whereby:

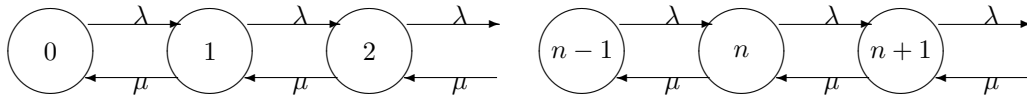
- The first letter indicates the nature of the arrival process
- The second letter indicates the nature of the pdf of the service time
- The last number indicates the number of servers

The letter “M” stands for memoryless, which here means a Poisson process (i.e., exponentially distributed interarrival times), “G” stands for a general distribution of interarrival times, and “D” stands for deterministic interarrival times.

The M/M/1 queue is the simplest queue system and may be described by selecting the birth-death coefficients as follows:

$$\begin{aligned}\lambda_n &= \lambda, \quad n = 0, 1, 2, \dots \\ \mu_n &= \mu, \quad n = 1, 2, \dots\end{aligned}$$

The state transition diagram is shown below:



From the previous birth-death results, we have

$$P_n = \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1 \left(1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}\right)}, \quad n \geq 1 \quad (5.2)$$

and

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \frac{\lambda_{n-1}\lambda_{n-2}\cdots\lambda_1\lambda_0}{\mu_n\mu_{n-1}\cdots\mu_2\mu_1}} \quad (5.3)$$

For M/M/1, this becomes

$$P_n = P_0 \left(\frac{\lambda}{\mu}\right)^n \quad (5.4)$$

where

$$P_0 = \frac{1}{1 + \sum_{n=1}^{\infty} \left(\frac{\lambda}{\mu}\right)^n}$$

Since $\lambda < \mu$ to have an equilibrium, the P_0 becomes

$$P_0 = 1 - \frac{\lambda}{\mu}$$

Letting $\rho = \lambda/\mu$, we have

$$P_n = (1 - \rho)\rho^n, \quad n = 0, 1, 2, \dots \quad (5.5)$$

This is indeed the solution for steady-state probability of finding n customers in the system.

To find the average number of customers in the system $\bar{N}$,

$$\begin{aligned} \bar{N} &= \sum_{n=0}^{\infty} nP_n \\ &= (1 - \rho) \sum_{n=0}^{\infty} n\rho^n \\ &= (1 - \rho)\rho \frac{\partial}{\partial \rho} \sum_{n=0}^{\infty} \rho^n \\ &= (1 - \rho)\rho \frac{\partial}{\partial \rho} \frac{1}{1 - \rho} \\ &= \frac{\rho}{1 - \rho} \end{aligned} \quad (5.6)$$

We may now apply Little's result to obtain T , the average time in the system as follows:

$$\begin{aligned} T &= \frac{\bar{N}}{\lambda} \\ &= \frac{\rho}{1 - \rho} \cdot \frac{1}{\lambda} \\ &= \frac{1/\mu}{1 - \rho} \end{aligned} \quad (5.7)$$

Question: Find average number of customers in queue, average waiting time in queue.

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Ex 5.2.1 Packets arrive at a transmission link according to a Poisson process with rate 100 packets per sec. Let's assume that packet length is exponentially distributed with mean 1000 bytes and data rate of the transmission link is 10 Mbps. Find $\bar{N}$ and T .

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5.3 The M/M/m, M/M/∞, M/M/m/m Systems

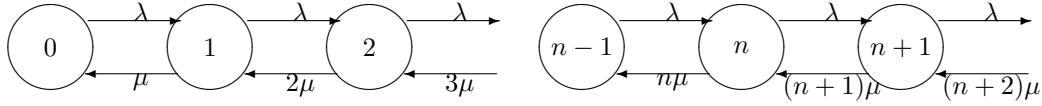
5.3.1 M/M/∞

In this system, we have

$$\lambda_n = \lambda, \quad n = 0, 1, 2, \dots$$

$$\mu_n = n\mu, \quad n = 1, 2, \dots$$

The state transition diagram is shown below:



From the general solution (5.2) of the birth-death process, we have

$$P_n = P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu} = P_0 \frac{(\lambda/\mu)^n}{n!} \quad (5.8)$$

where

$$\begin{aligned} P_0 &= 1 / \left(1 + \sum_{n=1}^{\infty} \frac{\lambda^n}{\mu^n n!} \right) \\ &= 1 / \left(\sum_{n=0}^{\infty} \frac{(\lambda/\mu)^n}{n!} \right) \\ &= 1 / e^{\lambda/\mu} \end{aligned}$$

Thus,

$$P_n = \frac{(\lambda/\mu)^n}{n!} e^{-\lambda/\mu}, \quad n = 0, 1, 2, \dots \quad (5.9)$$

From this (having the form of Poisson distribution), we have

$$\bar{N} = \frac{\lambda}{\mu} \quad (5.10)$$

Little's result provides

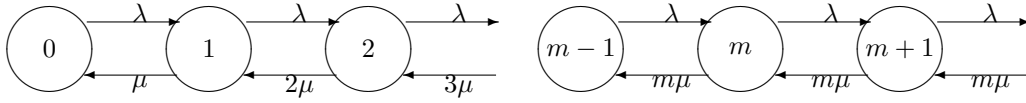
$$T = \frac{1}{\mu} \quad (5.11)$$

5.3.2 M/M/m

We consider a system with unlimited waiting room and with a constant arrival rate λ . The system provides for a maximum m servers. Since this is still a birth-death process, we have

$$\begin{aligned}\lambda_n &= \lambda, \quad n = 0, 1, 2, \dots \\ \mu_n &= \min(n\mu, m\mu) \\ &= \begin{cases} n\mu, & 0 \leq n < m \\ m\mu, & n \geq m \end{cases}\end{aligned}$$

The state transition diagram is shown below:



From (5.2) we have, for $n < m$,

$$\begin{aligned}P_n &= P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu} \\ &= P_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!}\end{aligned}\tag{5.12}$$

Similarly, for $n \geq m$,

$$\begin{aligned}P_n &= P_0 \prod_{i=0}^{m-1} \frac{\lambda}{(i+1)\mu} \prod_{j=m}^{n-1} \frac{\lambda}{m\mu} \\ &= P_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{m!m^{n-m}}\end{aligned}\tag{5.13}$$

Combining these results, we have

$$P_n = \begin{cases} P_0 \frac{(m\rho)^n}{n!}, & n < m \\ P_0 \frac{\rho^n m^m}{m!}, & n \geq m \end{cases}\tag{5.14}$$

where $\rho = \lambda/(m\mu) < 1$. To find P_0 we use (5.3):

$$\begin{aligned}P_0 &= \left(1 + \sum_{k=1}^{m-1} \frac{(m\rho)^k}{k!} + \sum_{k=m}^{\infty} \frac{(m\rho)^k}{m!} \frac{1}{m^{k-m}}\right)^{-1} \\ &= \left(\sum_{k=0}^{m-1} \frac{(m\rho)^k}{k!} + \frac{(m\rho)^m}{m!} \frac{1}{1-\rho}\right)^{-1}\end{aligned}\tag{5.15}$$

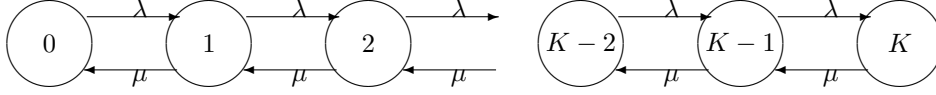
5.3.3 M/M/1/K

We now consider a queueing system in which there is a maximum number of customers that may be stored; in particular, we assume the system can hold at most a total of K customers and any further arriving customers will be in fact be refused entry to the system and will depart immediately without service.

Using the birth-death model,

$$\begin{aligned}\lambda_n &= \begin{cases} \lambda, & n < K \\ 0, & n \geq K \end{cases} \\ \mu_n &= \mu, \quad n = 1, 2, \dots, K\end{aligned}$$

The state transition diagram is shown below:



From (5.2), we have

$$\begin{aligned}P_n &= P_0 \prod_{i=0}^{n-1} \frac{\lambda}{\mu} \\ &= P_0 \left(\frac{\lambda}{\mu} \right)^n, \quad n \leq K\end{aligned}\tag{5.16}$$

For $n > K$,

$$P_n = 0, \quad n > K$$

To find P_0 , we use these in (5.3)

$$\begin{aligned}P_0 &= \left(1 + \sum_{k=1}^K \left(\frac{\lambda}{\mu} \right)^k \right)^{-1} \\ &= \left(1 + \frac{(\lambda/\mu)(1 - (\lambda/\mu)^K)}{1 - \lambda/\mu} \right)^{-1} \\ &= \frac{1 - \lambda/\mu}{1 - (\lambda/\mu)^{K+1}}\end{aligned}$$

Thus, finally,

$$P_n = \begin{cases} \frac{1 - \lambda/\mu}{1 - (\lambda/\mu)^{K+1}} \left(\frac{\lambda}{\mu} \right)^n, & 0 \leq n \leq K \\ 0, & \text{Otherwise} \end{cases}\tag{5.17}$$

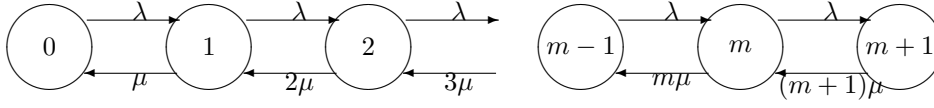
5.3.4 M/M/m/m

We consider a queueing system in which with m servers a new customer is given a private server, but the customer is lost when all servers are occupied.

Following the birth-death process, we have

$$\begin{aligned}\lambda_n &= \begin{cases} \lambda, & n < m \\ 0, & n \geq m \end{cases} \\ \mu_n &= n\mu, \quad n = 1, 2, \dots, m\end{aligned}$$

The state transition diagram is shown below:



Applying (5.2) we obtain

$$P_n = P_0 \prod_{i=0}^{n-1} \frac{\lambda}{(i+1)\mu}, \quad n \leq m$$

or

$$P_n = \begin{cases} P_0 \left(\frac{\lambda}{\mu}\right)^n \frac{1}{n!}, & n \leq m \\ 0, & n > m \end{cases} \quad (5.18)$$

Solving for P_0 we have

$$P_0 = \left(\sum_{k=0}^m \left(\frac{\lambda}{\mu}\right)^k \frac{1}{k!} \right)^{-1}$$

Specifically, P_m describes the fraction of time that all m servers are busy. The name given to this probability is Erlang's loss formula and it is given by

$$P_m = \frac{(\lambda/\mu)^m / m!}{\sum_{k=0}^m (\lambda/\mu)^k / k!} \quad (5.19)$$

This equation is also referred to as Erlang's B formula.

Ex 5.3.1 How many users can be supported for 0.5% blocking probability for the following number of trunked channels in a blocked calls cleared system? (a) 5, (b) 10. Assume each user generates 0.1 Erlangs of traffic.

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Note: The basic measure of traffic is the traffic intensity A , expressed in a dimensionless unit, the Erlang:

$$A = \lambda \times \frac{1}{\mu}$$

Erlang A can be interpreted in several ways:

- i. The average number of calls during the average holding period
- ii. The average number of channels required

Ex 5.3.2 Consider a cellular system where handoff calls are prioritized over originating calls by assigning a portion of channels exclusively for handoff calls. Let's assume that S_R channels are assigned exclusively for handoff calls among S channels in a cell. The remaining $S_C (= S - S_R)$ channels are shared by both calls. Find the blocking probability of originating call and handoff call. Assume call arrival and departure follows Poisson processes. And the arrival rates of originating and handoff calls are λ_o and λ_h , respectively, and the service rate is μ .

Sol:

From the local balance equations we have

$$\begin{aligned} n\mu P_n &= (\lambda_0 + \lambda_h)P_{n-1} & 0 \leq n \leq S_c \\ n\mu P_n &= \lambda_h P_{n-1} & S_c < n \leq S \end{aligned}$$

The steady-state probability is

$$P_n = \begin{cases} \frac{(\lambda_0 + \lambda_h)^n}{\mu^n n!} P_0, & 0 \leq n \leq S_c \\ \frac{(\lambda_0 + \lambda_h)^{S_c} \lambda_h^{n-S_c}}{\mu^n n!} P_0, & S_c < n \leq S \end{cases} \quad (5.20)$$

where

$$P_0 = \left(\sum_{i=0}^{S_c} \frac{(\lambda_0 + \lambda_h)^i}{\mu^i i!} + \sum_{i=S_c+1}^S \frac{(\lambda_0 + \lambda_h)^{S_c} \lambda_h^{i-S_c}}{\mu^i i!} \right)^{-1}$$

Let P_b and P_d denote blocking probability and handoff dropping probability. Then these are given by

$$P_b = \sum_{n=S_c}^S P_n$$

and

$$P_d = P_S = \frac{(\lambda_0 + \lambda_h)^{S_c} \lambda_h^{S-S_c}}{\mu^S S!} P_0$$

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